INTRODUCTION TO ★ ★ COMPUTER SCIENCE

The basic idea:

Input → ALGORITHMS → Output

Step by step instructions for solving problems.

The Computer Language: Computers use the "BINARY" Notation.

→ Humans use the numbers 1-9 (Base 10 nos.) while computers use 0 and 1 (Base 2 nos.)

↓ Bits ⇒ Binary digits

★ The only physical input to our devices is "electricity."

↳ We harness this electricity and try to represent information with it.

Metaphor of light:

off

0

on

1

- With one light bulb there are only two possibilities
- With three light bulbs, we can count till 7.

(Assuming you start counting from 0)

→ Inside a computer, there are no bulbs, but Switches which represent '1' for 'on' and '0' for 'off' are present. These are present in millions.

Now,

100	10	1
1	2	3

10^2	10^1	10^0
#	#	#

This is the mathematical system (Base 10). Here,

$1 = 10^0$ $\quad$ $100 = 10^2$

$10 = 10^1$

Also,

2^2	2^1	2^0
#	#	#

This is the binary system (Base 2). Here, we have the ones, twos and fours place.

Eg.)

0 = 000

1 = 001

2 = 010

3 = 011

⋮

so on. - - - -

Q) What decimal number does binary number 110010 represent.

A)

2^5	2^4	2^3	2^2	2^1	2^0
1	1	0	0	1	0

$2^5 + 2^4 + 2^1 \Rightarrow 32 + 16 + 2 \Rightarrow \boxed{50}$

American standard code for information interchange.

ASCII VALUES

We can assign specific number to other characters such as the english alphabet.

Eg) 'A' = 65 = 01000001

8 digits → Represent 8 bits → 1 Byte

- 1 Nibble → 4 bits
- 1 Byte → 8 bits
- 1 Kilobyte → 1024 bytes
- 1 Megabyte → 1024 KB
- 1 Gigabyte → 1024 MB
- 1 Terabyte → 1024 GB
- 1 Petabyte → 1024 TB
- 1 Exabyte → 1024 PB
- 1 Zettabyte → 1024 EB
- 1 Yottabyte → 1024 ZB

- Q) How many symbols can you represent with 8 bits?
- A) 256

2 2 2 2 2 2 2 2 → 256

UNICODE

Represent all symbols + emojis too.

Eg.) Emoji → Face with tears of joy.
→ 128514 → 0000000 11111 0 11 000000010

→ We need more than 256 characters.
↳ We can use 16, 24, 32 etc. bits.

COLOR REPRESENTATION

- All colors are essentially made up of RGB (Red, Green and Blue). But there are other acronyms.

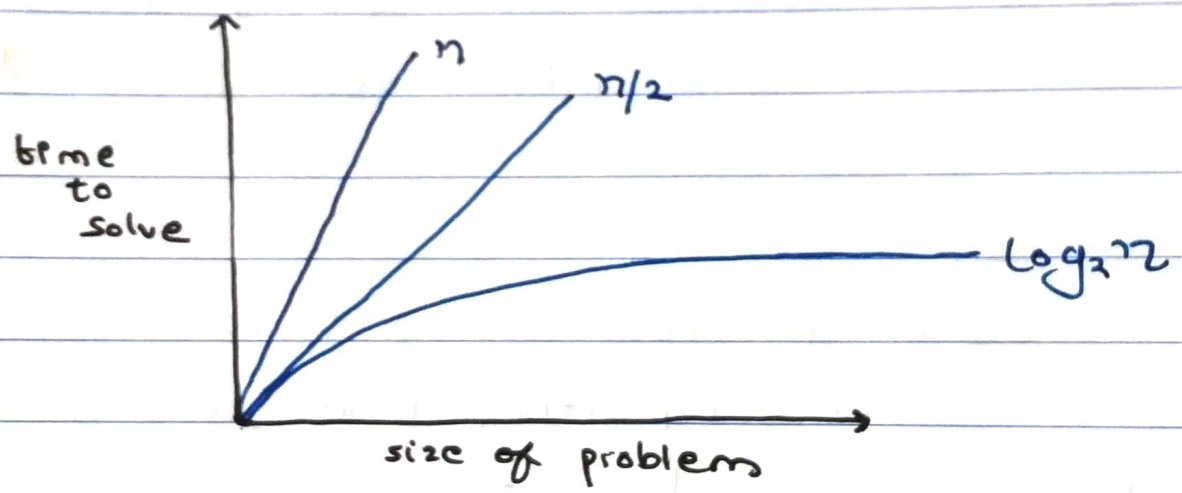
8 bits	+	8 bits	+	8 bits
0 - 255		0 - 255		0 - 255
R		G		B

- Computers use these to control the colours of pixels.

VIDEO FILES

- The frames change multiple times per second which looks like a video.
- They are rapidly changing pixels.
- In the world of computers too, the principle of a flipbook is applied.

⇒ Solving a problem efficiently:



⇒ PSEUDOCODE: Algorithm implemented without code.

Eg:)

1. Pick up phone book
2. open to the middle of phone book
3. look at page

etc.